Group-Aware Splits for Educational Sequence Data: How Student Overlap Inflates Reported Accuracy

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Abstract

Educational datasets are made of many short sequences from relatively few students. When windows or attempts are split into train and test sets at random, the same student appears on both sides and models can learn the student rather than the behavior. We re-ran the evaluations of eleven prediction tasks on five public-style datasets with random splits and with splits grouped by student. Grouped splits lowered reported performance by 4 to 17 points of AUC or F1, with the largest drops for tasks built from short windows. We give a checklist for detecting and reporting this leakage.

1 The problem

Suppose each student contributes 40 windows and a model sees 32 of them in training. Typing speed, preferred variable names and habitual error types identify the student almost as well as a name would. If the label distribution differs by student, a model can score well by recognizing who wrote a window.

This is a known form of leakage, but it is easy to introduce: most libraries split rows, not groups, by default.

2 Re-evaluation

We re-implemented eleven tasks (dropout, struggle, help-seeking, error type and others) on five datasets and evaluated each with random 5-fold splits and with 5-fold splits grouped by student.

Task family	Unit	Random split	Grouped split	Drop
Struggle detection	30 s window	0.83 F1	0.69 F1	14
Help-seeking	60 s window	0.79 F1	0.71 F1	8
Next error type	attempt	0.74 F1	0.65 F1	9
Dropout	student-week	0.81 AUC	0.77 AUC	4
Off-task	30 s window	0.71 F1	0.54 F1	17

Drops were largest for short windows and for labels that were unevenly distributed across students. Dropout prediction, already defined per student-week, changed least.

3 Detecting leakage

A quick diagnostic: train a model to predict the student id from the same features. If it is far above chance, random splits are unsafe. On our struggle-detection data, student id was predicted with 64% top-1 accuracy among 120 students.

4 Checklist

- 1. State the unit of the split (row, attempt, student, section, term).
- 2. Group by student unless the claim is explicitly about known students.

• 3. Report the student-id probe accuracy.

• 4. If prior numbers used random splits, report both.

5 Limitations

Grouped splits answer a different question: performance on new students. Some applications, such as personalization for an enrolled student, may legitimately use within-student data.

References

[1] Kaufman, S., Rosset, S., Perlich, C., and Stitelman, O. Leakage in data mining: formulation, detection, and avoidance. ACM TKDD 6, 4 (2012), Article 15.